

27	C	In Fig. (a), for sinusoidal a.c., $V_{r.m.s. \text{ for (a)}} = \frac{V_P}{\sqrt{2}}$ In Fig. (b), for the case of full square wave ac, $V_{peak} = V_{r.m.s.}$, the mean power is $\langle P \rangle_{full} = \frac{V_{Peak}^2}{R}$ But Fig. (b) represents a half-wave rectified square wave, $P_{half} = \frac{V_{Peak}^2}{2R}$ $\frac{V_{r.m.s. \text{ for (b)}}^2}{R} = \frac{V_{Peak}^2}{2R} \Rightarrow V_{r.m.s. \text{ for (b)}} = \frac{V_P}{\sqrt{2}}$ Thus, the ratio $\frac{V_{r.m.s. \text{ for (a)}}}{V_{r.m.s. \text{ for (b)}}} = \frac{\frac{V_P}{\sqrt{2}}}{\frac{V_P}{\sqrt{2}}} = 1$
28	A	By Photoelectric Equation: $hf = \phi + KE$